

10(a)	packet / quantum of energy	M1
	of electromagnetic radiation	A1
10(b)(i)	light is re-emitted in all directions / only part of the re-emitted light is in the direction of the beam	B1
10(b)(ii)	an arrow between -3.40 eV and -1.51 eV <u>and</u> an arrow between -3.40 eV and -0.85 eV	B1
	all arrows shown point 'upwards'	B1
10(b)(iii)	$E = hc / \lambda$ or $E = hf$ <u>and</u> $c = f\lambda$	C1
	$2.60 \times 1.60 \times 10^{-19} = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$	C1
	$\lambda = 4.8 \times 10^{-7} \text{ m}$	A1

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Question	Answer	Marks
11	any five from: • electrons need energy to enter conduction band (from valence band) • (positively-charged) holes are left in valence band • moving charge carriers/holes/electrons are current • (increase of temperature leads to) more (positive and negative) charge carriers/more holes/more electrons so more current • more charge carriers/holes/electrons gives rise to less resistance • (increase of temperature causes) greater (amplitude of) vibrations of atoms/ions/lattice • effect of more charge carriers/holes/electrons is greater than effect of greater vibrations (and so resistance decreases)	B5